

Smart Personal Finance & Goal Management System

Project Proposal for UCS503P

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1 Higher-order goal

... secondary framing

This project aims to improve personal financial decision-making by helping users not only track their income and expenses, but also understand their financial behavior and take concrete actions toward long-term goals. Most personal finance applications primarily present financial information through balances, transactions, and charts. However, users often struggle to translate this information into consistent financial behavior.

The higher-order goal of this project is therefore to build a **goal-oriented personal finance system** that converts financial data into actionable recommendations, supports planning under uncertainty, and helps users make informed decisions while maintaining privacy and transparency.

2 Time-to-value

... primary heuristic

- We will deliver meaningful functionality incrementally, beginning with secure transaction tracking, financial goals, budgeting, and a basic dashboard.
- We will prioritize rapid feedback by developing the system in iterations and validating each major feature through automated tests and user-oriented scenarios.
- Advanced features such as behavioral pattern detection, anomaly detection, scenario simulation, and shared-finance workflows will be added incrementally after the core financial model is stable.
- CI/CD automation will be introduced early so that every significant change can be tested consistently and deployed in a repeatable manner.

3 Problem Statement

Many personal finance applications focus primarily on displaying financial information such as balances, transactions, spending categories, and monthly summaries. While these features are useful for understanding past spending, they do not necessarily help users change their financial behavior or determine whether they are on track to achieve future goals.

Common challenges include:

- **Limited goal orientation:** users can see their spending but may not understand how their current behavior affects goals such as building an emergency fund, purchasing a laptop, planning a trip, or saving for a major expense.
- **Irregular income and expenses:** traditional monthly budgets often assume predictable income and expenses and therefore provide limited guidance when financial conditions vary.
- **Weak behavioral feedback:** users may repeatedly exhibit spending patterns without receiving specific, timely, and actionable suggestions.
- **Shared financial complexity:** households may have multiple contributors, shared expenses, different ownership arrangements, and approval requirements that are difficult to represent using simple expense-splitting mechanisms.
- **Limited planning under uncertainty:** a single projection may give users a false sense of certainty when future income, expenses, and savings behavior can change.
- **Lack of transparent intervention:** financial applications may recommend actions without clearly explaining why a recommendation was generated or allowing users to inspect and modify the underlying rules.

Why this matters now (contextual relevance): Users increasingly manage multiple financial goals and irregular expenses while making decisions under uncertainty. A system that connects financial behavior with future goals can provide more useful guidance than a system that only reports historical transactions.

4 Proposed Features

The proposed system will provide the following major features:

- **Goal-based budgeting:** Users can create financial goals, allocate budgets toward them, and track their progress over time.
- **Scenario-based financial planning:** The system will simulate different income, expense, and savings scenarios to estimate the likelihood of achieving a financial goal.
- **Behavior and habit analysis:** Transaction history will be analyzed to identify recurring spending patterns and provide personalized suggestions for improving financial habits.
- **Shared household finances:** Multiple users will be able to manage shared expenses with clear ownership, contribution tracking, approval flows, and fairness metrics.
- **Anomaly detection and interventions:** The system will detect unusual or rapidly increasing spending patterns and provide gentle, actionable recommendations.
- **Explainable rule engine:** Users will be able to inspect and modify the rules used to generate financial recommendations.
- **Privacy and security:** Financial information will be protected using secure data handling, encryption, least-privilege access, and audit logs.
- **Audit and scenario comparison:** Users will be able to export financial records and compare different financial scenarios before making major decisions.

5 Proposed Solution

... *Software Proposition*

5.1 Overview

We propose a **web-based Smart Personal Finance & Goal Management System** that combines financial tracking, goal-based budgeting, behavioral analysis, scenario simulation, and actionable interventions.

The major components are:

- **Financial tracking:** users record and categorize income, expenses, accounts, and transactions.
- **Goal management:** users define financial goals with target amounts, deadlines, priorities, and progress.
- **Goal-based budgeting:** the system connects spending and savings behavior with financial goals and provides recommendations for staying on track.
- **Scenario simulation:** users can compare different financial scenarios and use Monte-Carlo-style simulations to estimate the likelihood of achieving a goal under varying income, expenses, and savings conditions.
- **Habit and transaction analysis:** the system detects recurring spending patterns and links them to user-defined or automatically suggested financial rules.
- **Anomaly detection and intervention:** unusual or rapidly increasing spending patterns are identified and converted into gentle, concrete recommendations rather than only displaying warnings.
- **Shared household finances:** users can manage shared expenses, ownership, approval flows, contributions, and fairness metrics.
- **Audit and decision support:** users can export financial records and compare scenarios for significant financial decisions.

5.2 Core workflow

1. A user securely creates an account and records or imports financial transactions.
2. Transactions are categorized and associated with relevant accounts, goals, budgets, or household members.
3. The system analyzes historical financial behavior to identify recurring patterns and unusual changes.
4. The user creates a financial goal by specifying a target amount, deadline, and other relevant preferences.
5. The system estimates whether the goal is achievable using current financial behavior and generates alternative scenarios.
6. The user can modify assumptions such as income, expenses, or monthly savings and compare the resulting scenarios.
7. When the system detects a potentially important spending pattern or deviation, it provides an explainable intervention with a concrete suggested adjustment.
8. Users can inspect, modify, accept, or reject the rules behind automated recommendations.

5.3 Operational constraints for a privacy-first financial system

- Financial information must be protected through secure authentication, authorization, encryption, and least-privilege access.
- The system should minimize collection of sensitive information and store only the data required for its intended functionality.
- Recommendations should be explainable and should not make irreversible financial decisions automatically.
- The system will provide financial planning and visualization features without attempting to function as a full brokerage platform.
- Money calculations must be implemented carefully and tested extensively to avoid rounding and arithmetic errors.

6 Solution Approach

... Engineering focus

This is an engineering course project, so the focus will be on reliable system design, secure handling of financial information, explainable decision support, testing, and maintainability rather than purely research-oriented algorithmic novelty.

6.1 Goal-based budgeting and scenario simulation

The system will connect current financial behavior with future goals. Instead of assuming a single deterministic future, the system will allow users to compare multiple scenarios.

Monte-Carlo-style simulation will be used to generate multiple possible future financial paths based on assumptions about income, expenses, savings, and other relevant variables. The resulting distribution can be used to estimate the likelihood of achieving a financial goal by a specified deadline.

Users will be able to compare scenarios such as:

- maintaining current spending behavior,
- reducing selected discretionary expenses,
- increasing monthly savings,
- experiencing lower or irregular income, and
- changing the target date or goal amount.

6.2 Habit and transaction rule engine

The system will analyze transaction history to identify recurring or meaningful spending patterns. Based on these patterns, it can suggest rules that users can inspect and modify.

For example, repeated spending in a particular category may result in a suggestion to set a monthly limit or redirect a portion of future savings toward a selected goal.

The rule engine will be designed to remain transparent so that users can understand why a recommendation was generated and modify or disable it when required.

6.3 Anomaly detection and intervention

The system will identify unusual spending behavior using historical transaction patterns and category-level trends.

Instead of presenting only an “anomaly detected” message, the system will provide a gentle intervention containing:

- the category or behavior that changed,
- an explanation of the deviation,
- the potential impact on relevant goals, and
- a concrete suggested adjustment.

6.4 Shared household finance workflow

The system will support multiple users participating in shared financial activities. Each shared transaction can have ownership information and, where appropriate, an approval state.

The system will maintain contribution information and provide fairness metrics based on configurable sharing rules rather than assuming that all expenses must always be divided equally.

6.5 Web architecture

... web-based solution

A typical 3-tier architecture will be used:

- **Frontend:** responsive user interface for dashboards, transactions, budgets, goals, scenarios, and household workflows.
- **Backend API:** authentication, authorization, transaction processing, budgeting logic, rule execution, anomaly detection, scenario simulation, and audit logging.
- **Database:** storage for users, accounts, transactions, categories, goals, budgets, rules, scenarios, household relationships, approvals, and audit records.

6.6 Security and auditability

Financial information will be treated as sensitive application data. The system will incorporate:

- authentication and role-based authorization,
- least-privilege access,
- encryption for sensitive data and secure communication,
- secure secret management,
- validation of financial inputs, and
- append-oriented audit logs for important financial actions.

6.7 CI/CD and testing

CI/CD will be used to automatically build the application, run automated tests, perform static checks, and produce repeatable deployment artifacts.

Special attention will be given to financial calculations. Unit tests, integration tests, and property-based tests will be used to validate important properties of money calculations, budgeting logic, scenario calculations, and goal projections.

7 Evaluation Criterion

... measurable and attributable

Success will be evaluated using measurable criteria related to correctness, goal planning, recommendation quality, and system reliability.

7.1 Primary evaluation metric

Goal projection accuracy and consistency: the system should produce consistent and mathematically valid goal projections when identical assumptions are supplied.

The evaluation will compare simulated outcomes against controlled test scenarios with known expected behavior.

7.2 Secondary evaluation metrics

- **Goal achievement feasibility:** percentage of test scenarios in which the system correctly identifies whether a goal is feasible under the specified assumptions.
- **Recommendation usefulness:** user evaluation of whether suggested interventions are understandable, relevant, and actionable.
- **Anomaly detection quality:** ability to identify meaningful deviations without producing excessive false alerts.
- **Calculation correctness:** correctness of balances, budgets, contributions, projections, and other money-related calculations.
- **System reliability:** successful completion of critical workflows such as authentication, transaction recording, budgeting, and goal creation.
- **Security and auditability:** verification that unauthorized users cannot access protected financial information and that important financial actions generate appropriate audit records.

7.3 Pilot validation plan

... high-level

A small group of users will be given representative financial scenarios and asked to complete common tasks such as creating a goal, categorizing transactions, comparing scenarios, and responding to spending interventions.

The evaluation will combine automated test results with user feedback on clarity, usefulness, and ease of understanding.

8 Scalability

... with foresight

The system should be scalable both technically and functionally.

1. **Scalable architecture:** the frontend, backend services, and database will be separated so that individual components can evolve independently.
2. **Database scalability:** indexes, pagination, efficient queries, and appropriate data modelling will be used for transaction-heavy workloads.
3. **Time-series scalability:** transaction history and financial events will be stored in a form suitable for historical analysis and forecasting.
4. **Simulation scalability:** scenario simulations can be separated from regular request processing when computational requirements increase.

5. **Operational scalability:** CI/CD and containerized or platform-hosted deployment can be used to support repeatable deployments.

9 Engine availability heuristic

A mature set of software engineering tools and technologies is available to implement the proposed system:

- Web framework for the frontend and backend application.
- Relational database for structured financial and user data.
- Authentication and authorization mechanisms for secure access.
- Time-series processing and forecasting libraries for financial history and scenario analysis.
- Statistical and numerical libraries for Monte-Carlo-style simulations.
- Automated testing frameworks including support for property-based testing.
- CI/CD tools for automated testing and deployment.

These mature technologies will allow the project to focus on financial workflow design, correctness, privacy, explainability, and user behavior rather than building infrastructure from scratch.

10 Project scope and deliverables

... iteration-friendly

10.1 Initial deliverable

... first iteration

- Secure user authentication and basic authorization.
- Income and expense transaction management.
- Transaction categorization and financial dashboard.
- Creation and tracking of financial goals.
- Basic goal-based budgeting.
- Initial database schema and secure backend API.
- Automated unit and integration tests for core financial calculations.
- CI pipeline for building and testing the application.

10.2 Subsequent deliverables

- Habit and transaction pattern detection.
- User-inspectable rule engine for financial recommendations.
- Anomaly detection and gentle intervention system.
- Monte-Carlo-style goal projections and scenario comparison.
- Shared household finances with ownership and approval workflows.

- Fairness metrics for shared financial contributions.
- Tax-aware and investment-aware financial views without brokerage functionality.
- Exportable audit trail and decision scenario reports.
- Property-based testing for important financial calculations.

11 Risks and mitigations

- **Financial data privacy:** minimize sensitive data collection, enforce least-privilege access, use encryption, and maintain secure audit logs.
- **Incorrect financial calculations:** use precise numerical representations, clearly defined calculation rules, extensive automated testing, and property-based tests.
- **Unhelpful recommendations:** keep recommendations explainable, allow users to inspect and modify rules, and avoid automatically making irreversible financial decisions.
- **False anomaly alerts:** use historical patterns and configurable thresholds while allowing users to dismiss or provide feedback on alerts.
- **Simulation uncertainty:** clearly communicate that scenario projections are estimates based on assumptions rather than guaranteed outcomes.
- **Scope complexity:** implement the core tracking, goal, and budgeting functionality first and introduce advanced features in subsequent iterations.
- **Security vulnerabilities:** use secure coding practices, dependency checks, automated testing, and restricted access to sensitive resources.

12 Summary

This project proposes a privacy-first Smart Personal Finance & Goal Management System designed to move beyond passive financial tracking toward goal-oriented decision support and behavioral change. The system will combine transaction management, goal-based budgeting, behavioral pattern detection, anomaly-based interventions, scenario simulation, shared household finances, and an inspectable rule engine.

The project emphasizes software engineering depth through secure financial data handling, scalable architecture, time-series analysis, forecasting, Monte-Carlo-style simulation, explainable recommendations, auditability, and strong automated testing of money-related calculations.

By connecting everyday financial behavior with long-term goals and showing users the potential impact of alternative decisions, the system aims to help users understand not only *where their money went*, but also *what they can do next* to improve their financial outcomes.